

# High-dimensional Covariance Matrix Estimation

Heqing Shi

University of Edinburgh Business School, Edinburgh, UK  
Heqing.Shi@ed.ac.uk  
<http://shiheqing.github.io>

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## 1 Issues With High-dimensional Data

Consider a matrix  $\mathbf{X}_{10 \times 3}$  with each column representing a feature of a dataset

$$\mathbf{X} = \begin{pmatrix} 0.3 & 0.85 & 1.2 \\ 0.2 & -0.06 & 0.3 \\ \vdots & \vdots & \vdots \\ 0.35 & -0.7 & 0.05 \end{pmatrix}. \quad (1)$$

Such a system is well-tractable. We are able to direct estimate the covariance matrix  $\Sigma_{3 \times 3}$  for the three features efficiently using the sample covariance

$$\hat{\Sigma} = \mathbf{s}, \quad (2)$$

without too much estimation error. Since we have 3 diagonal elements plus three off-diagonal elements (6 parameters in total) to estimate given 30 observations.

Sever issues occurs when the dimensionality of the data becomes larger.....

Say, we have  $N$  different features with  $T$  observations. When  $N \gg T$ , we will have in total  $\frac{N(N+1)}{2}$  free parameters to estimate, given only  $T \times N$  observations!

$$\mathbf{X}_{T \times N} = \begin{pmatrix} x_{11} & \dots & x_{1N} \\ \vdots & \ddots & \vdots \\ x_{T1} & \dots & x_{TN} \end{pmatrix} \quad (3)$$

$$\mathbf{s}_{N \times N} = \begin{pmatrix} \sigma_1^2 & \dots & \sigma_{1N} \\ \vdots & \ddots & \vdots \\ \sigma_{N1} & \dots & \sigma_N^2 \end{pmatrix} \quad (4)$$

If we insist to use the sample covariance as the estimator of  $\Sigma_{N \times N}$ , then we will encounter the following issues:

- singular  $\mathbf{s}_{N \times N}$  for sure  $\rightarrow$  not invertible!
- unacceptable estimation errors, particularly in off-diagonal entries
- any estimations based on  $\mathbf{s}_{N \times N}$  will be problematic

The high-dimensional issues for covariance matrix estimation is common to see, especially in finance. Think about estimating a covariance matrix for a large cross-section of assets. When using historical data (usually monthly or up to daily), the number of assets  $N$  will be much larger than the periods of observed returns  $T$ .

## 2 Ledoit and Wolf (2003) Shrinkage Estimation

From statisticians' point of view, the problem of high-dimensional covariance matrix estimation can be cured by imposing structure to the covariance matrix. The covariance matrix with structure imposed though is probably biased (not asymptotically converge to the true covariance matrix as the sample covariance matrix does), but it comes with less estimation error.

The idea is straightforward, given a sample covariance matrix  $\mathbf{s}$  and a covariance matrix  $\mathbf{F}$  with a structure, we can “shrink”  $\mathbf{s}$  towards  $\mathbf{F}$  to obtain an improved covariance matrix estimation.

But how to estimate  $\mathbf{F}$ ?

Assume that the DGP is something close to Sharpe (1963) single-index model:

$$x_{it} = \alpha_i + \beta_i f_t + \epsilon_{it}, \quad (5)$$

where  $f_t$  is the dominant factor. The covariance matrix given such a DGP is

$$\tilde{\Sigma} = \mathbf{b} \text{Var}(f) \mathbf{b}^\top + \mathbf{\Delta}, \quad (6)$$

where  $\mathbf{\Delta}$  is a diagonal matrix with residual variances.

The model in (5) can be estimated using OLS.  $\mathbf{b}$ ,  $\mathbf{\Delta}$  can be obtained from the OLS result, and  $\text{Var}(f)$  can be estimated by the sample variance of  $f$ , which leads to

$$\mathbf{F} = \mathbf{b} s_f^2 \mathbf{b}^\top + \mathbf{D}. \quad (7)$$

Given the parsimonious structure of (5), the shrinkage target  $\mathbf{F}$  will have much lower estimation errors than the sample covariance matrix  $\mathbf{s}$  of the original dataset.

To regularize  $\mathbf{s}$ , we shrink it towards  $\mathbf{F}$  according to a linear convex combination:

$$\Sigma_{Shrink} = k\mathbf{F} + (1 - k)\mathbf{s}, \quad (8)$$

where  $k$  is the shrinkage constant. According to Ledoit and Wolf (2003), the optimal shrinkage constant follows  $k = (\pi - \rho)/\gamma$ . In Ledoit and Wolf (2003), the consistent estimators for all ingredients are also provided. For a clear notations, we can refer to a followed-up paper, Ledoit and Wolf (2004), which published in *Journal of Portfolio Management*.

There are many choices to impose a structure to a covariance matrix, which is literally an art. Ledoit and Wolf (2004) propose a constant correlation structure that performs comparable to the single-index structure. The structure tells that all pairwise correlations are the same, which can be estimated using the average of the sample correlations.

### 3 Fan et al. (2013) POET Estimation

This method starts from a similar rational with Ledoit and Wolf (2003), but is implemented in a very different routine. Let's also consider a structure (actually a factor model at this time):

$$x_{it} = \mathbf{b}_i^\top \mathbf{f}_t + \epsilon_{it}. \quad (9)$$

Now  $\mathbf{f}$  is not univariate, and we can assume a common  $K$ -factor structure. It can be shown that selecting a  $K$ -factor model is asymptotically equivalent to summarizing  $x_{it}$  with the first  $K$  principal components.

Same logic can be applied to regularize the sample covariance matrix  $\mathbf{s}$ . Since  $x_{it}$  is assumed to have common exposure on the first  $K$  principal factors  $\mathbf{f}$ , the covariance matrix of  $x_{it}$  then should have a sparser structure, i.e. the pairwise covariances are non-negligible among the first  $K$ -factors but not among the remaining.

Given a sample covariance matrix  $\mathbf{s}$ , we first do a spectral decomposition:

$$\mathbf{s} = \sum_{i=1}^K \hat{\lambda}_i \hat{\boldsymbol{\xi}}_i \hat{\boldsymbol{\xi}}_i^\top + \hat{\mathbf{R}}_K, \quad (10)$$

where  $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \hat{\lambda}_N$  are the ordered eigenvalues, and  $\{\hat{\boldsymbol{\xi}}_i\}_{i=1}^N$  are their corresponding eigenvectors, and  $\hat{\mathbf{R}}_K = \sum_{i=K+1}^N \hat{\lambda}_i \hat{\boldsymbol{\xi}}_i \hat{\boldsymbol{\xi}}_i^\top = (\hat{r}_{ij})_{N \times N}$  is the principal orthogonal complement.

Since the factors after the first  $K$  principals are less important, it is safe to assume that they will have tiny impact on the covariance matrix. We therefore apply thresholding on  $\hat{\mathbf{R}}$ , such as, according to

$$\hat{\mathbf{R}}_K^\mathcal{T} = (\hat{r}_{ij}^\mathcal{T})_{p \times p}, \quad \hat{r}_{ij}^\mathcal{T} = \begin{cases} \hat{r}_{ii}, & i = j, \\ s_{ij}(\hat{r}_{ij}) \mathbf{1}_{(|\hat{r}_{ij}| \geq \tau_{ij})}, & i \neq j, \end{cases} \quad (11)$$

How to choose the thresholding parameters is out of our scope in this discussion. For details we can refer to Fan et al. (2013), Section 2.2.

Plug in the thresholded principal orthogonal complement  $\hat{\mathbf{R}}_K^\mathcal{T}$  back to (10) gives us the POET estimator of the covariance matrix

$$\hat{\boldsymbol{\Sigma}}_K = \sum_{i=1}^K \hat{\lambda}_i \hat{\boldsymbol{\xi}}_i \hat{\boldsymbol{\xi}}_i^\top + \hat{\mathbf{R}}_K^\mathcal{T}. \quad (12)$$

## 4 Code Resources

For Ledoit and Wolf (2003) shrinkage estimation, we can use the scikit-learn implementation `sklearn.covariance.LedoitWolf()`. Check out here: <https://scikit-learn.org/stable/modules/generated/sklearn.covariance.LedoitWolf.html>. Alternatively, there is a written script in Python by others (<https://github.com/WLM1ke/LedoitWolf>). For more advanced implementation, you could code up yourself!

For Fan et al. (2013) POET estimation, there is an R package written by the original authors. Check out here: <https://cran.r-project.org/web/packages/POET/index.html>. I also find a Python script by others (<https://github.com/dizhouwu/POET>), which is convenient to use.

## Bibliography

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